# Code along: Writing functions and computing indices and segregation



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Rbanism

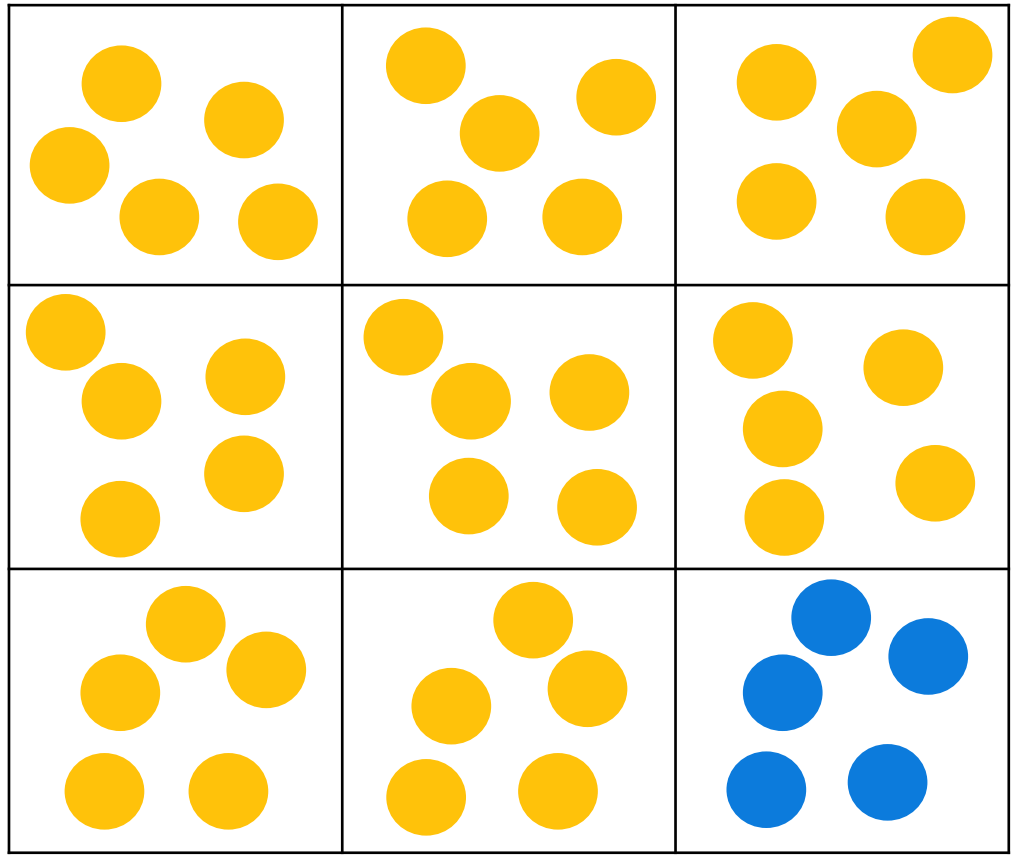
SEGUE

# Context

Most segregation indicators were originally designed to measure racial discrimination

# Context

- Let's imagine a city with 9 neighbourhoods with a total of 45 inhabitants, with a similar population density in every neighbourhood
- 40 inhabitants (89%) belong to **group *A***, and 5 (11%) to **group *B***
- If segregation were to be 0, each neighbourhood should have 89% of its residents belonging to group A and 11% to group B
- Segregation in this case can be calculated as the degree of deviation from this proportion in every neighbourhood



# Context

Let's take the case of the Index of Dissimilarity:

$$D = \frac{1}{2}\sum_{i=1}^{N}\left|\frac{a_i}{A} - \frac{b_i}{B}\right|$$

Where:

$a_i$ = the population of group *A* in the $i^{th}$ area

*A* = the total population in group *A* in the large geographic entity for which the index of dissimilarity is being calculated.

$b_i$ = the population of group *B* in the $i^{th}$ area

*B* = the total population in group *B* in the large geographic entity for which the index of dissimilarity is being calculated.

# Context

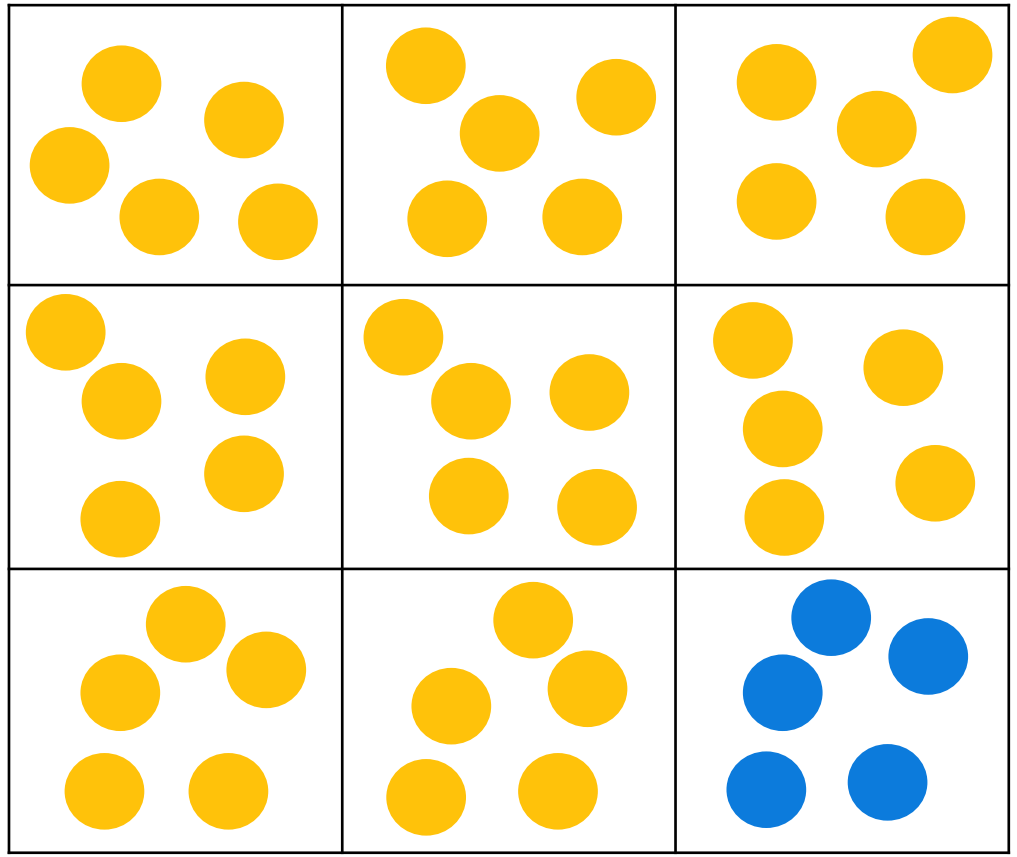
The formula for the DI:

$$D = \frac{1}{2}\sum_{i=1}^{N}\left|\frac{a_i}{A} - \frac{b_i}{B}\right|$$

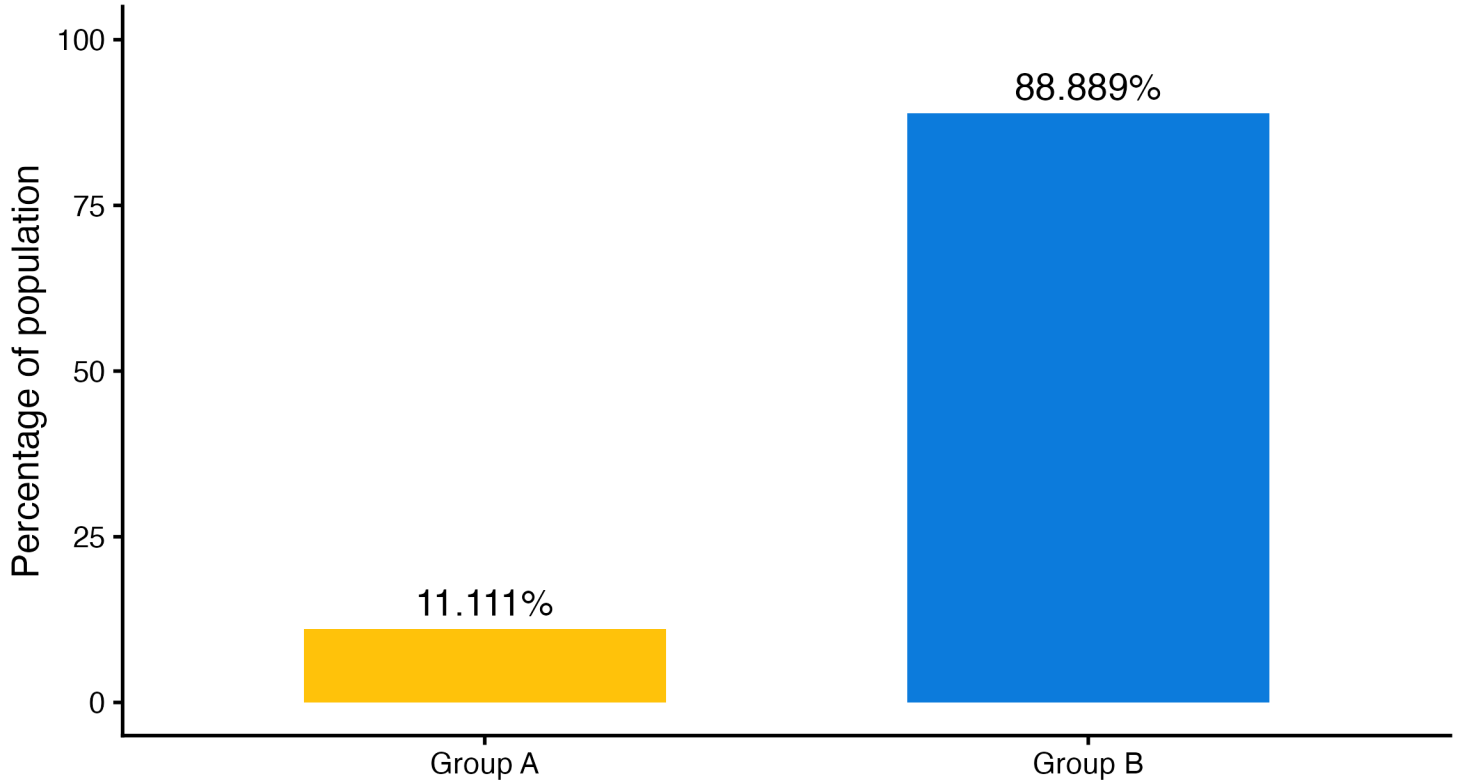
So…

$$D = \frac{12}{\left[8\left|\frac{5}{40} - \frac{0}{5}\right| + \left|\frac{0}{40} - \frac{5}{5}\right|\right]} = \frac{12}{(1+1)} = 1.$$

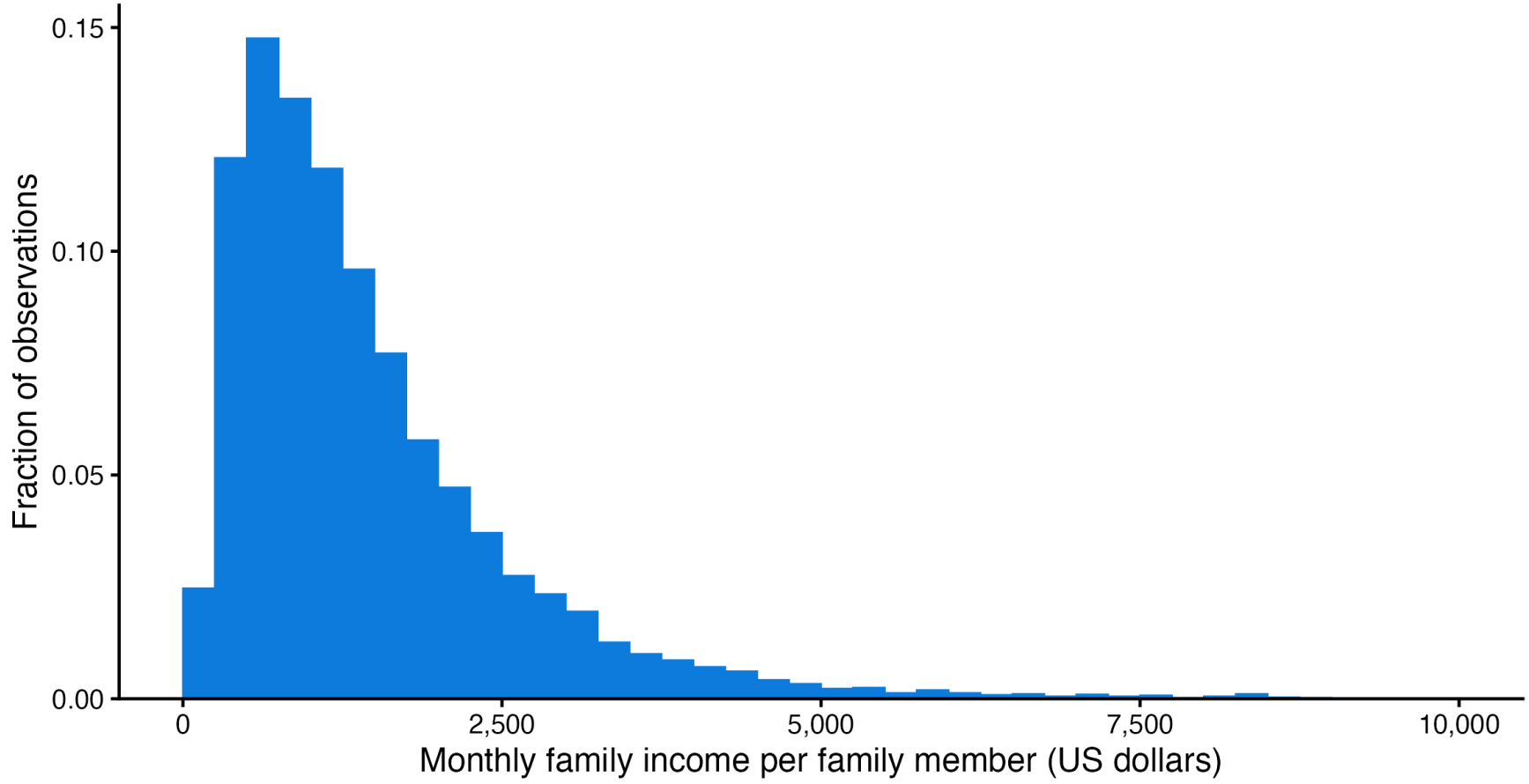
Our city registers the highest possible degree of segregation, with a **DI of 1**



# This works well when we deal with an underlying structure like...

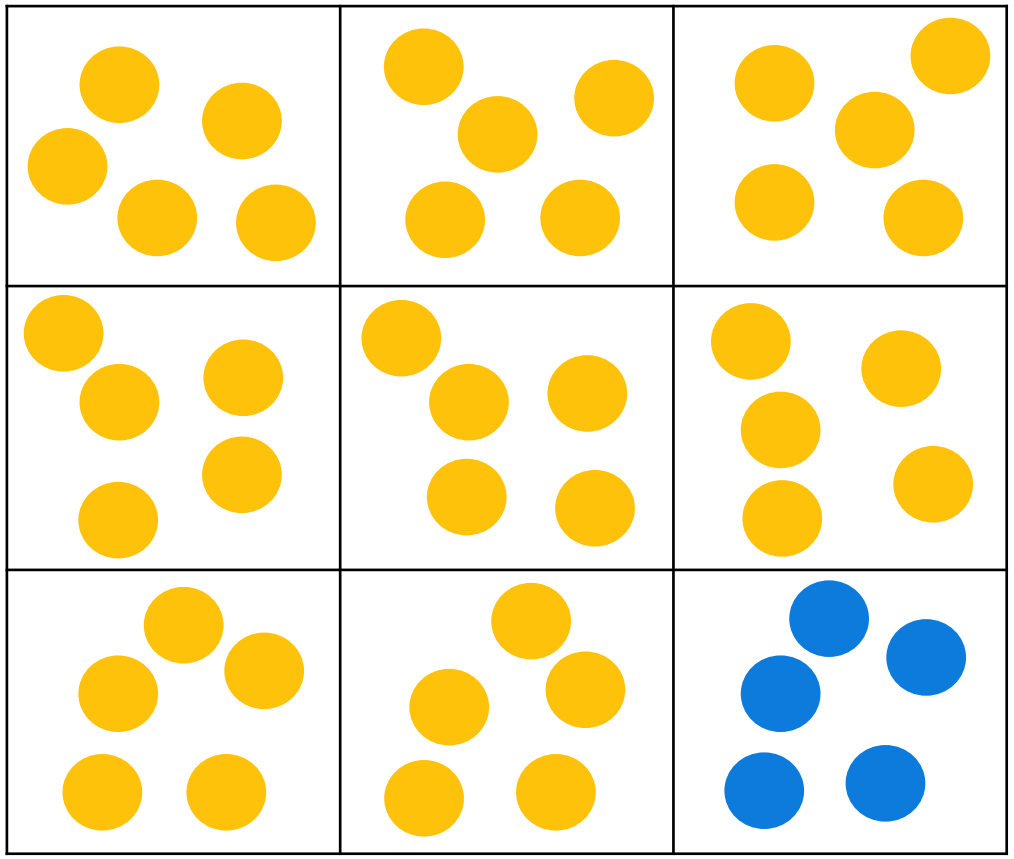


# But what happens when we deal with this?

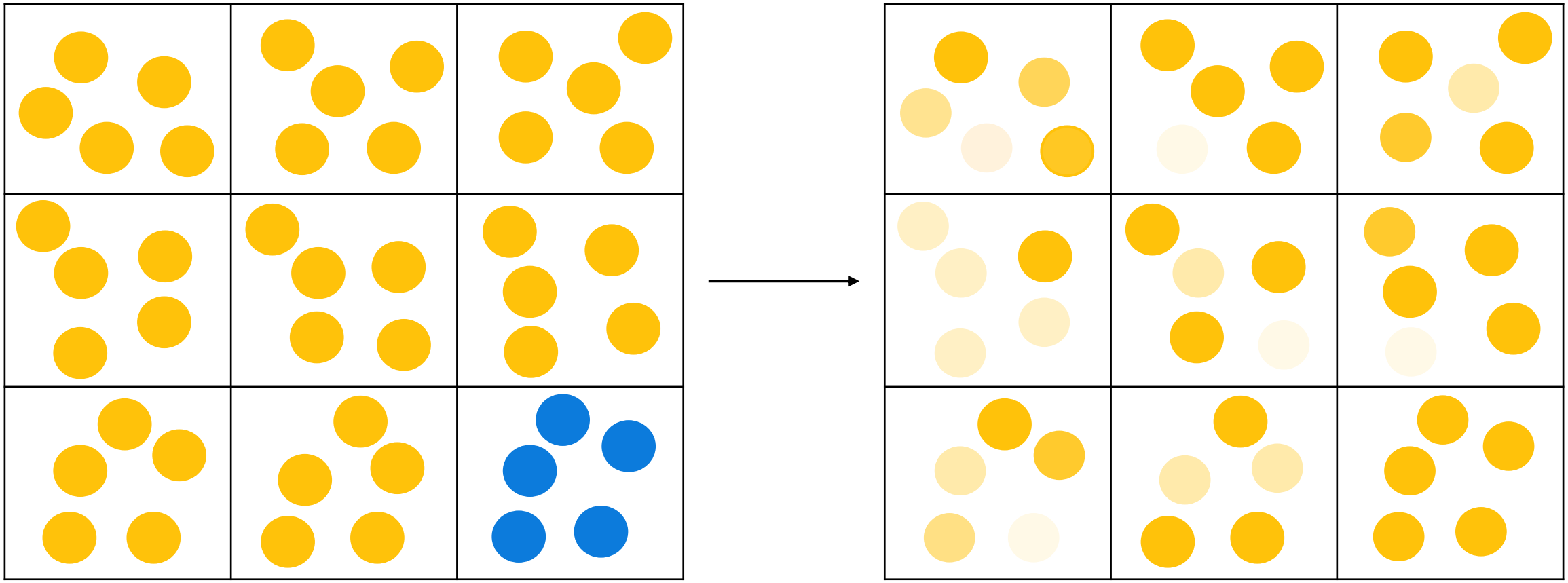



Source: https://vincentarelbundock.github.io/Rdatasets/csv/wooldridge/k401ksubs.csv

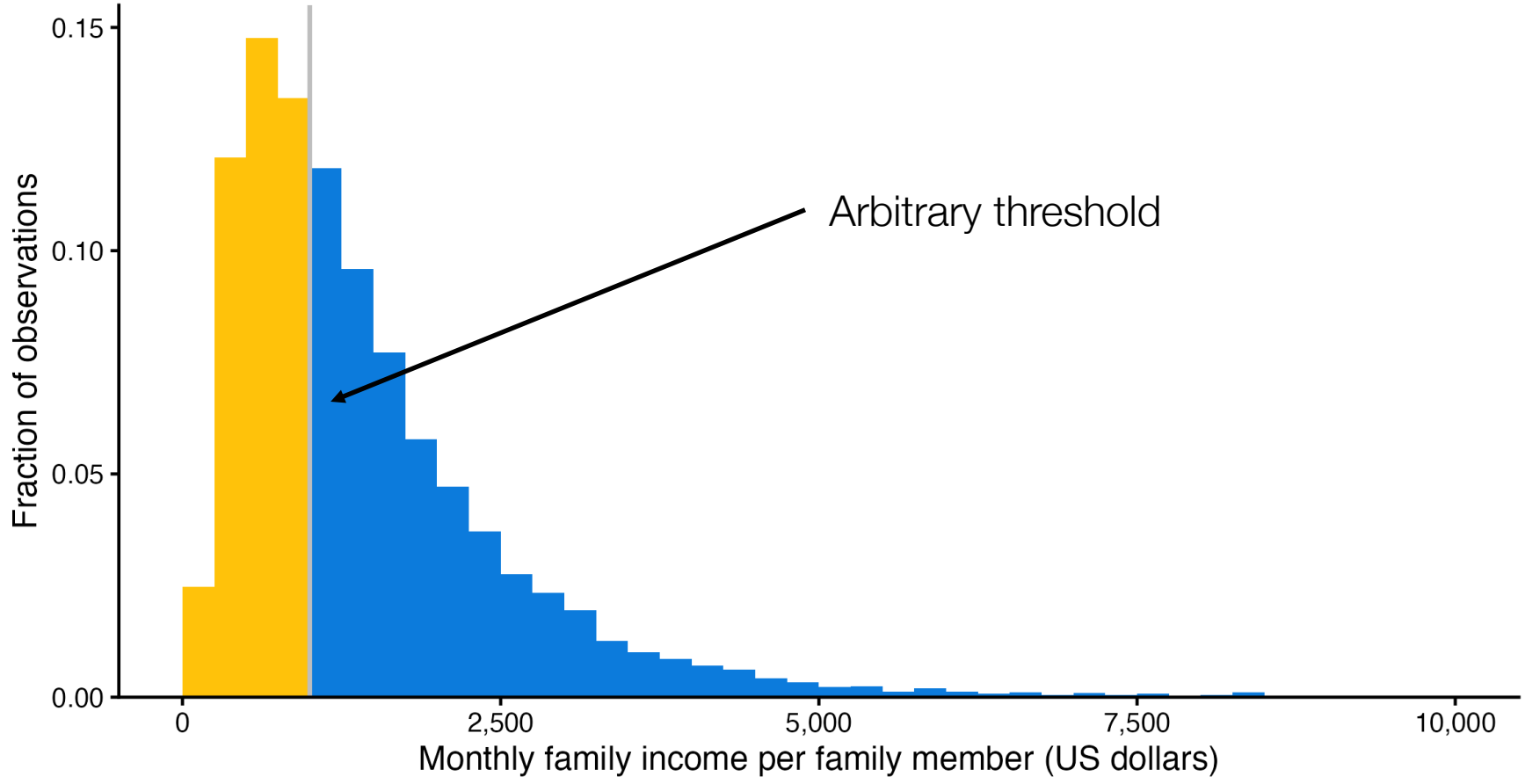
# This works well when we deal with an underlying structure like…



# But what happens when we deal with this?

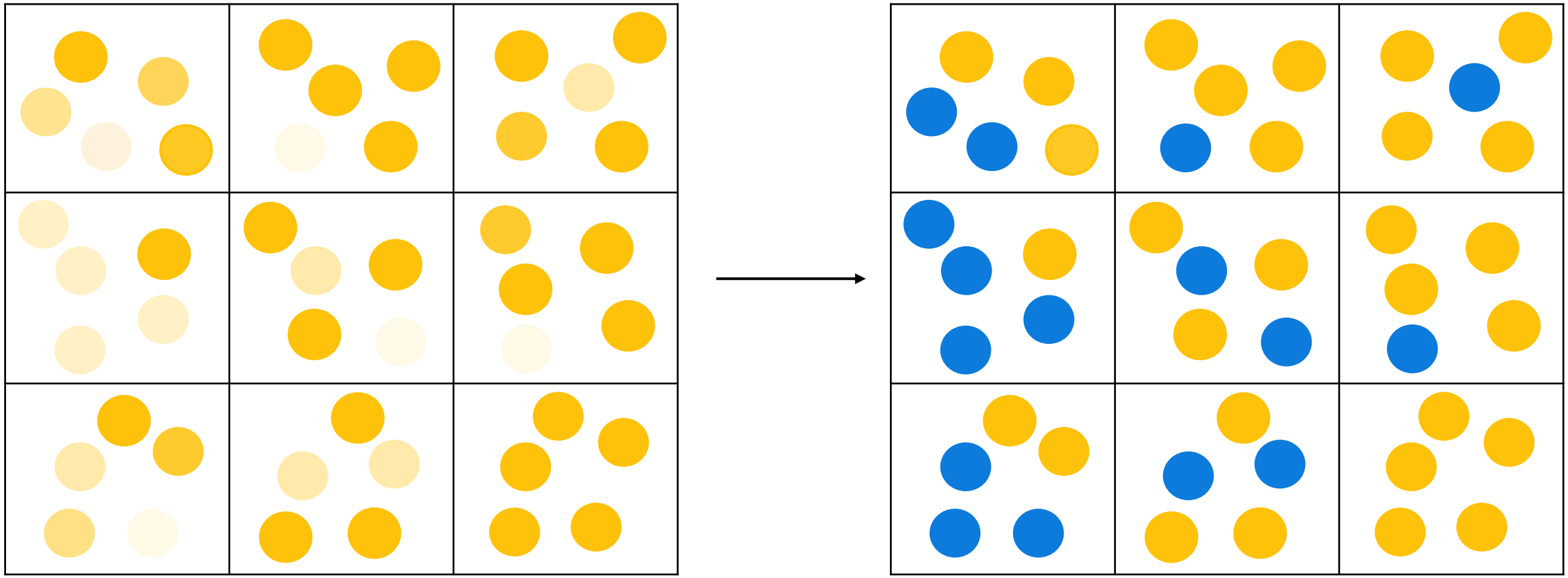


# Frequenty way to work with this…

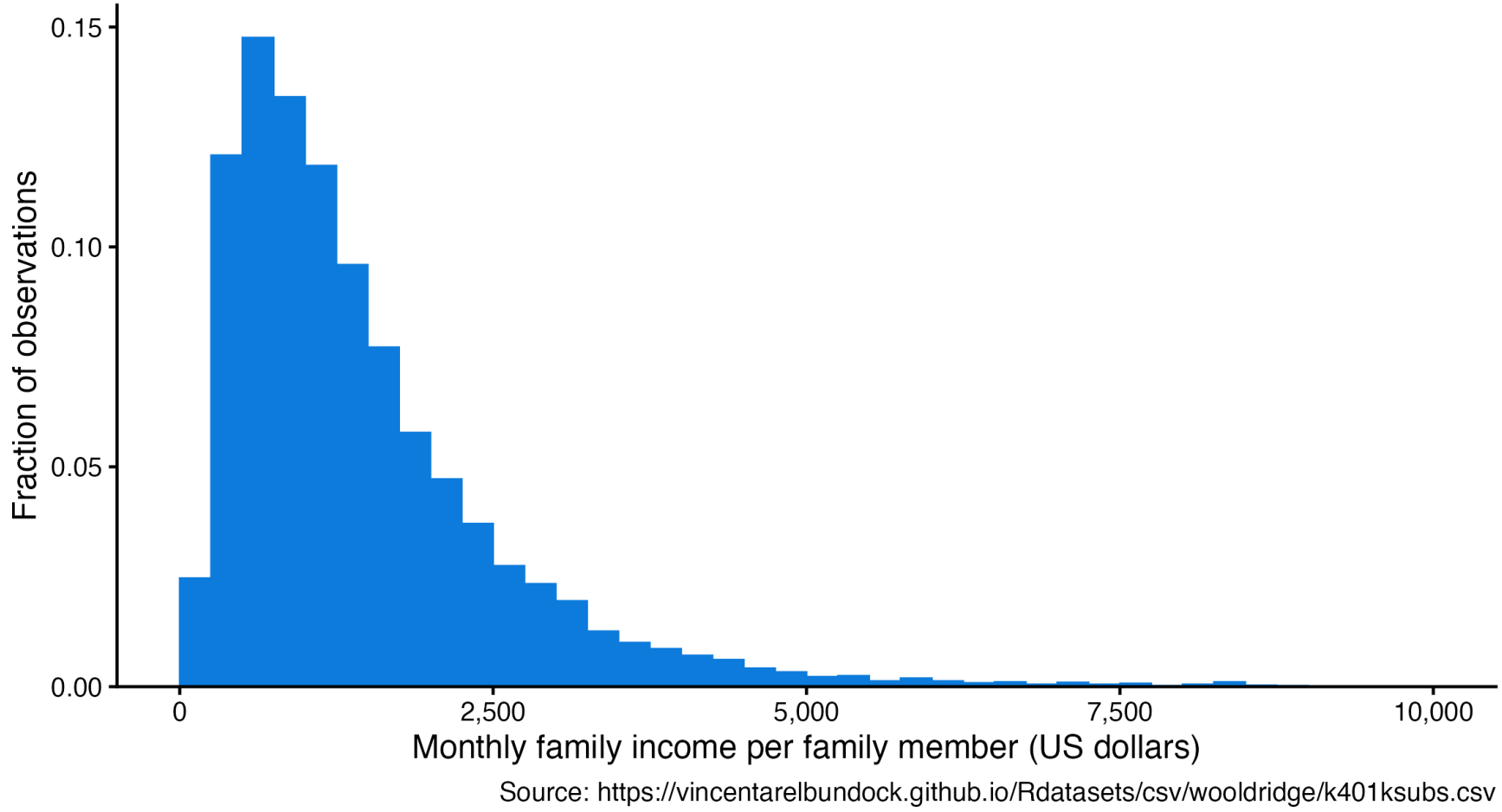



Source: https://vincentarelbundock.github.io/Rdatasets/csv/wooldridge/k401ksubs.csv

# Which is replicating the logic of racial segregation measurements



# And which does not represent well the underlying structure




Source: https://vincentarelbundock.github.io/Rdatasets/csv/wooldridge/k401ksubs.csv

# Measurements of economic segregation must consider:

The continuous nature of the variables related to economic segregation (i.e. income, wealth)

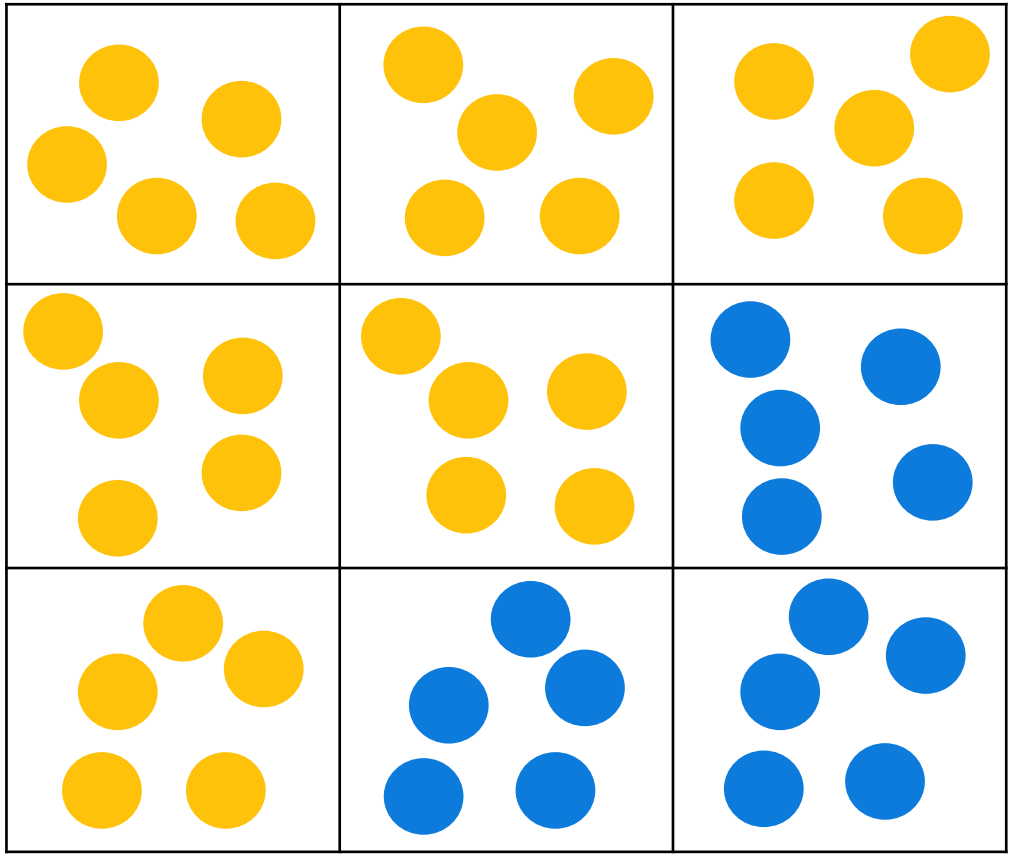
*Traditional indicators are designed to measure categories (e.g. racial discrimination, ethnic discrimination)*

# Are these two cities equally segregated?

**City 1 – 45 inhabitants**

Group A (2/3 of the population) + Group B (1/3 of the population)

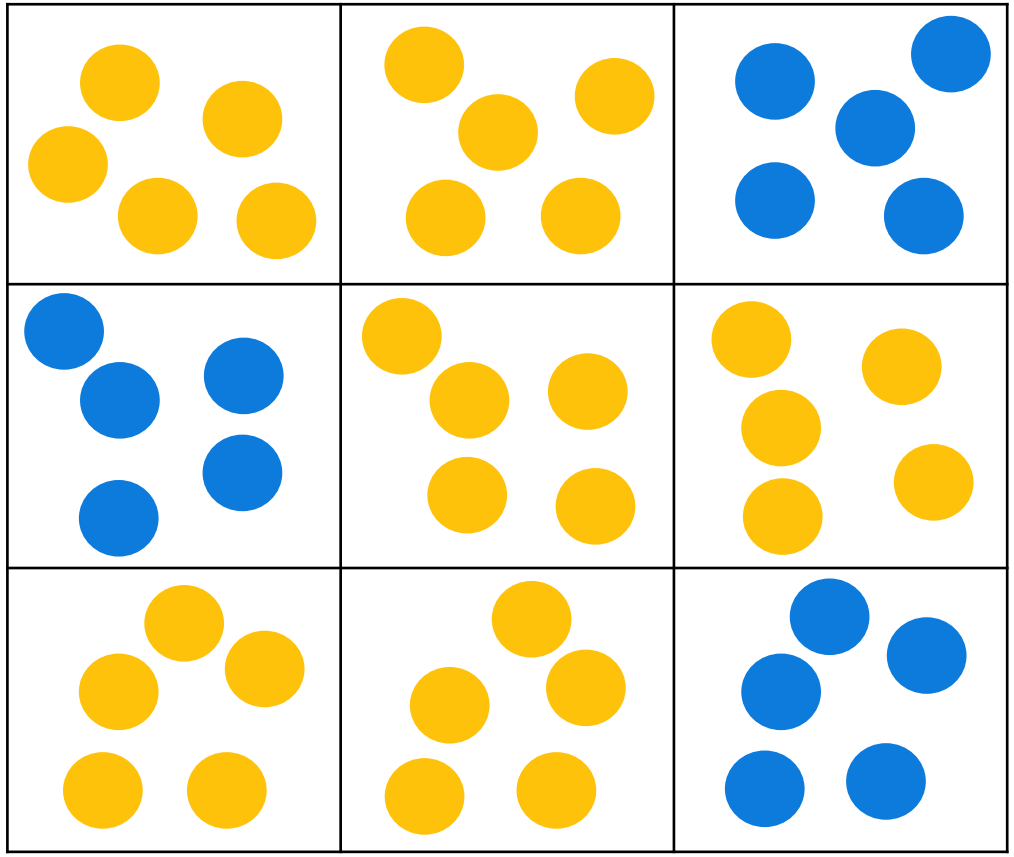
**Dissimilarity Index: 1**



**City 2 – 45 inhabitants**

Group A (2/3 of the population) + Group B (1/3 of the population)

**Dissimilarity Index: 1**



# Measurements of economic segregation must consider:

The continuous nature of the variables related to economic segregation (i.e. income, wealth)

→ *Traditional indicators are designed to measure categories (e.g. racial discrimination, ethnic discrimination)*

Space

→ *Segregation can happen in large-scale patterns i.e. district-wide segregation*

→ *Segregation can happen in a small-scale pattern i.e. mosaic-like segregation*

# For the our session, we propose computing an indicator that takes into account both elements

# Our choice

**The Spatial Rank-Ordered Information Theory Index (SROITI)**

- Adapted to a **continuous/ordinal** variable
- Does **not rely on arbitrary thresholds** and takes advantage of all income information
- Independent of the levels of economic disparities
- Adapted as a **spatial version**

The Rank-Order Information Theory Index

Reardon et al. (2006) describe the rank-order information theory index in detail; we summarize its key features here. First, let $p$ denote income percentile ranks (scaled to range from zero to one) in a given income distribution (i.e., $p = F(Y)$, where $Y$ measures income and $F$ is the cumulative income density function). Now, for any given value of $p$, we can dichotomize the income distribution at $p$ and compute the residential (pairwise) segregation between those with income ranks less than $p$ and those with income ranks greater than or equal to $p$. Let $H(p)$ denote the value of the traditional information theory index (Theil and Finezza 1971; Theil 1972; Zoloth 1976; James and Taeuber 1985) of segregation computed between the two groups so defined. Likewise, let $E(p)$ denote the entropy of the population when divided into these two groups (Theil and Finezza 1971; Theil 1972; Pielou 1977). That is,

$$E(p) = p \log_2 \frac{1}{p} + (1 - p) \log_2 \frac{1}{1 - p} \quad (1)$$

and

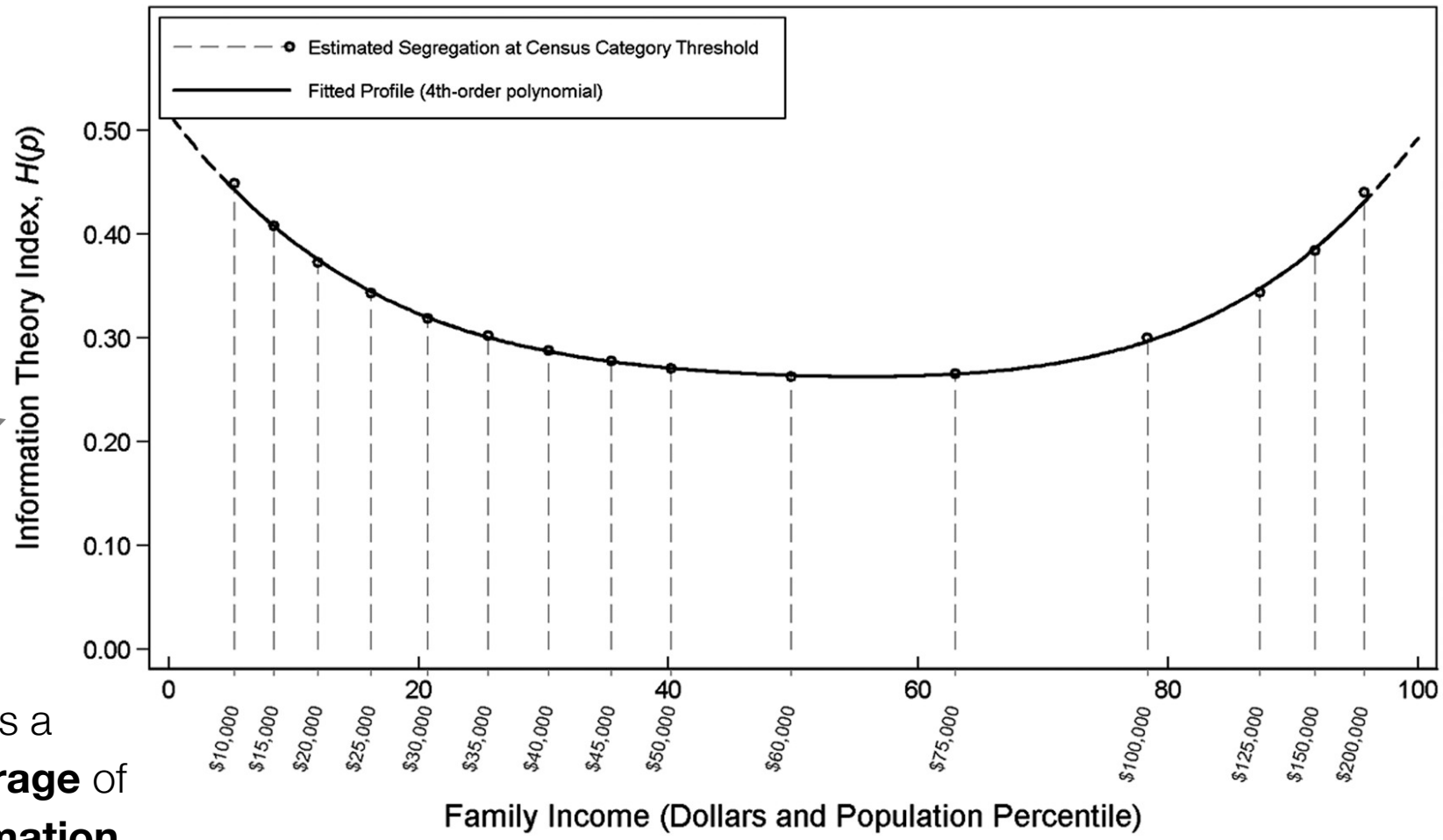
$$H(p) = 1 - \sum_j \frac{t_j E_j(p)}{TE(p)}, \quad (2)$$

where $T$ is the population of the metropolitan area and $t_j$ is the population of neighborhood $j$. Then the rank-order information theory index ($H^R$) can be written as

$$H^R = 2 \ln(2) \int_0^1 E(p) H(p) dp. \quad (3)$$

Extracted from: Reardon, S. F., & Bischoff, K. (2011). Income inequality and income segregation. *American Journal of Sociology, 116*(4), 1092-1153.

# How is it computed?

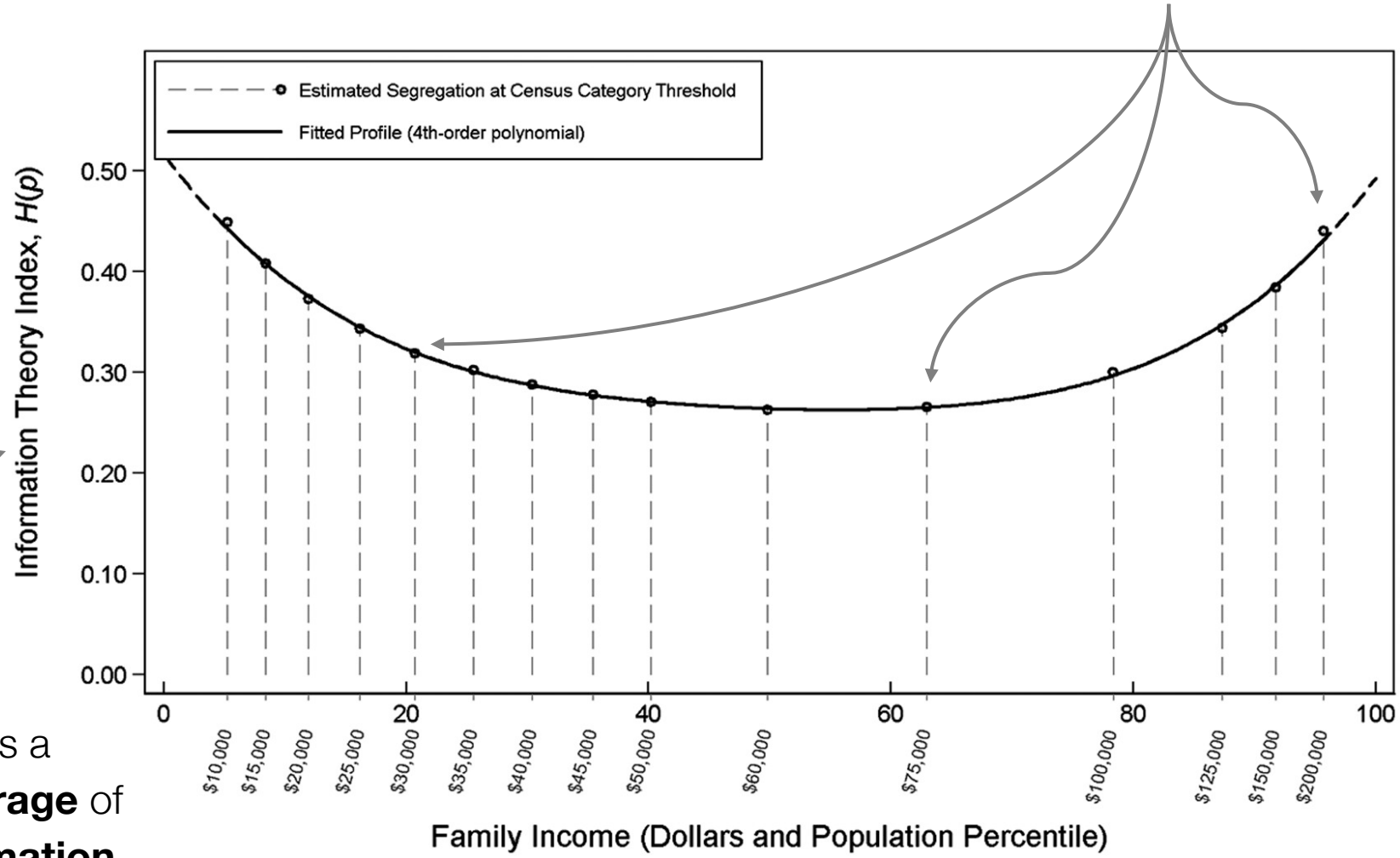



The ROITI is a **weighted average** of **several Information Theory Indexes** (ITI)

Extracted from: Reardon, S. F., & Bischoff, K. (2011). Income inequality and income segregation. *American Journal of Sociology, 116*(4), 1092-1153.

# How is it computed?

Each ITI represents the **segregation at a specific percentile** threshold




The ROITI is a **weighted average** of **several Information Theory Indexes** (ITI)

Extracted from: Reardon, S. F., & Bischoff, K. (2011). Income inequality and income segregation. *American Journal of Sociology, 116*(4), 1092-1153.

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Problem: existing R packages either…

- Compute spatial indicators
- Calculate indicators adapted to economic segregation

**But not both things at the same time**

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Problem: existing R packages either…

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**But not both things at the same time**

**There is no package with a function to compute the spatial version of the ROITI**

# In our session we plan to compute the spatial version of the Rank-Ordered Information Theory Index (SROITI)

**Using and combining existing packages** (i.e. *seg*) that calculate the spatial version of the Information Theory Index (ITI)

Creating **new *ad hoc* functions** that calculate a weighted average of ITIs, which gives us the SROITI

With open access, geo-coded (at the census tract level) income data (*Atlas de Distribución de la Renta de los Hogares*) from Spain

https://erc-segue.nl/2026Workshop/CodeAlong/Functions-and-Indices.html